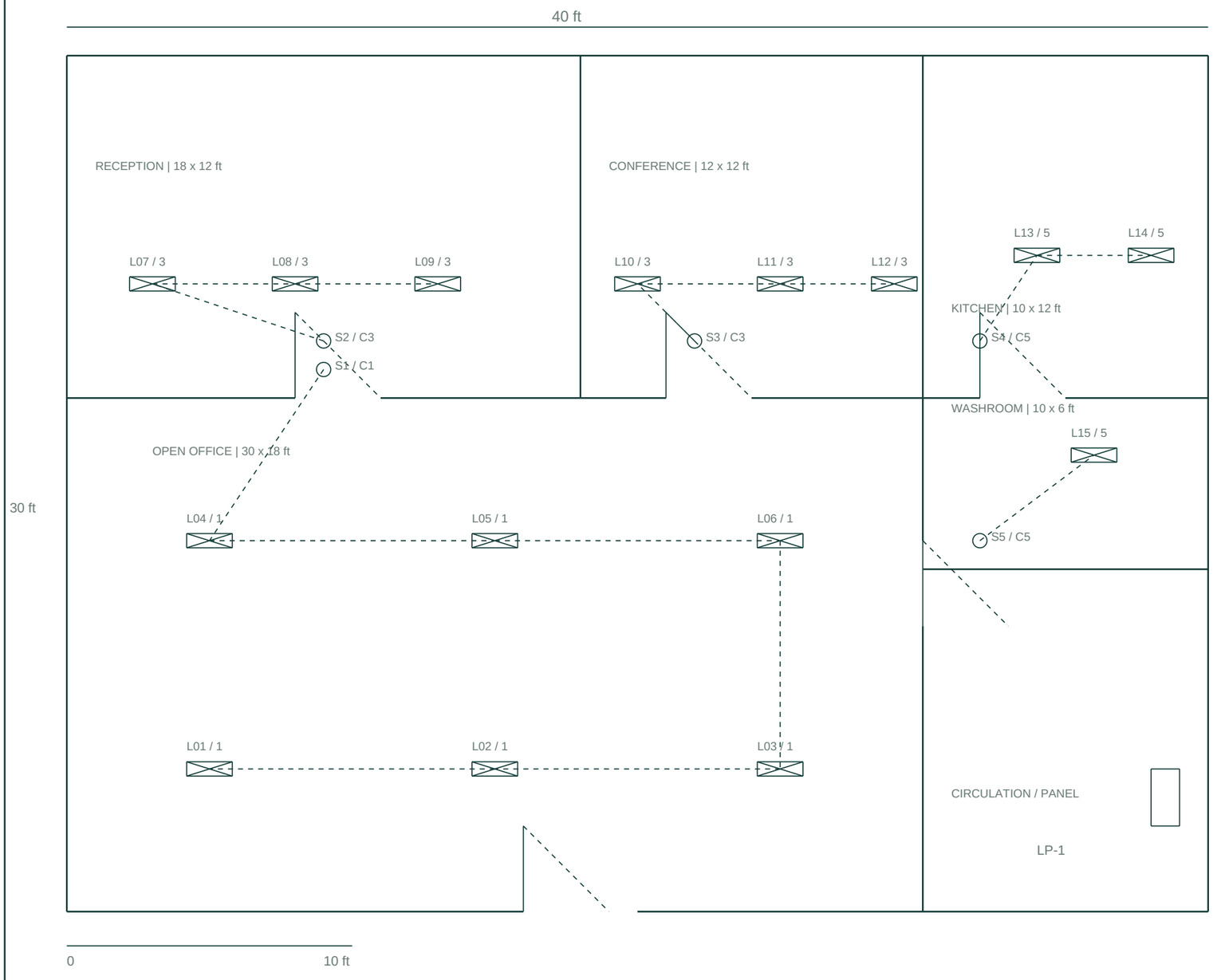
LIGHTING & SWITCHING PLAN

AMARACHI | COMMERCIAL OFFICE STUDY | 1,200 SQ. FT. | REV 01 | 2026-11-06



MODEL: 1 unit = 1 ft | Print scale 17 pt/ft at 17 x 11 in | Dimensioned geometry governs

DESIGN BASIS

15 illustrative LED fixtures at 40 VA each.

Lighting connected load: 600 VA.

C1: open office, 6 fixtures, 240 VA.

C3: reception + conference, 6, 240 VA.

C5: kitchen + washroom, 3, 120 VA.

Dashed paths show control associations, not cable routes or conduit quantities.

S1: open office; S2: reception; S3: conference; S4: kitchen; S5: washroom.

TO VERIFY

Fixture photometry, task illuminance, glare, ceiling coordination and energy controls.

Emergency lighting, exit signs and egress requirements are not designed in this set.

Switch locations / accessibility to be reviewed.

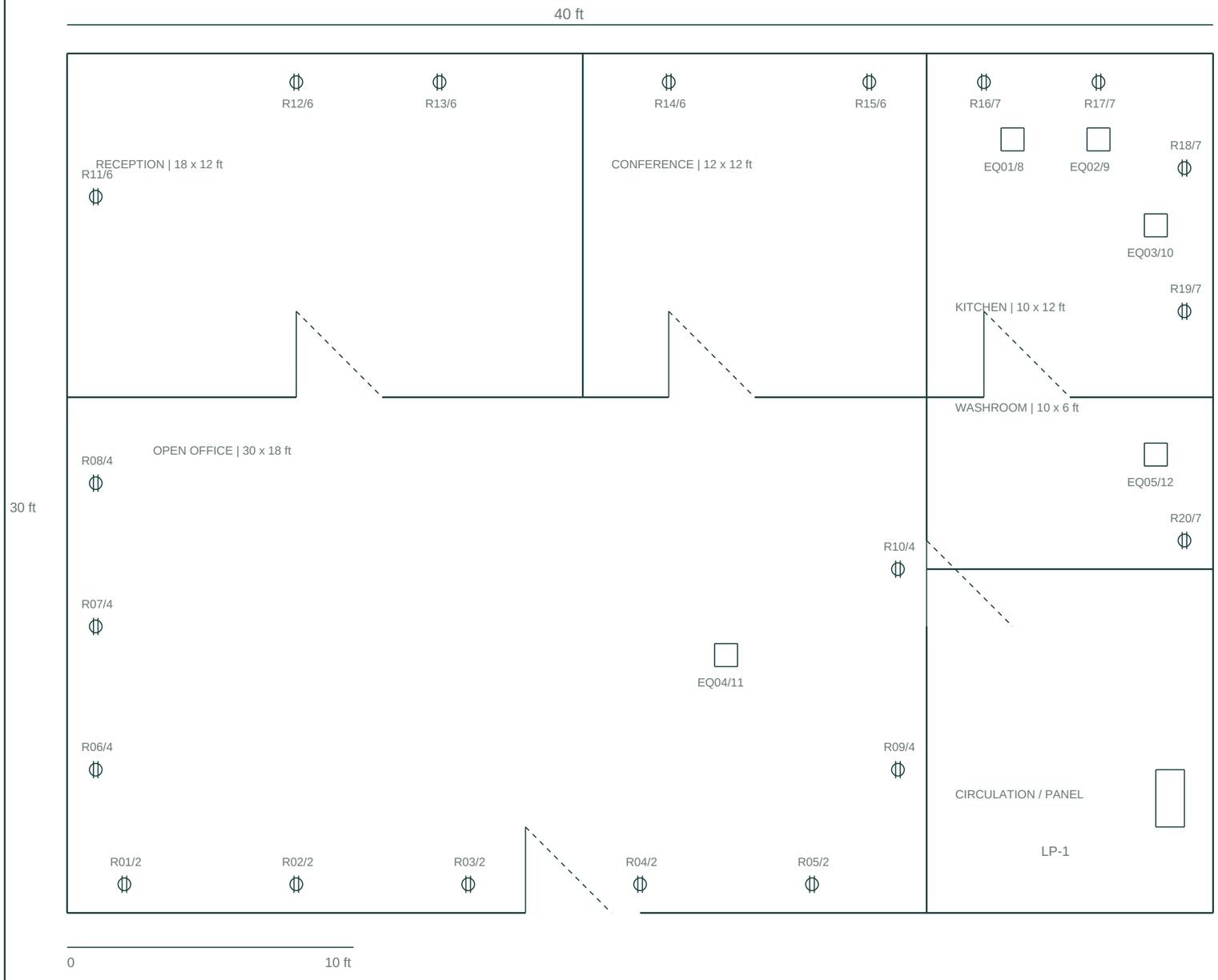
CONCEPTUAL DESIGN - NOT FOR CONSTRUCTION

Python-generated portfolio study. Not reviewed in AutoCAD; no code compliance, professional seal, or permit suitability claimed.

E-101

POWER & CIRCUIT IDENTIFICATION PLAN

AMARACHI | COMMERCIAL OFFICE STUDY | 1,200 SQ. FT. | REV 01 | 2026-11-06



MODEL: 1 unit = 1 ft | Print scale 17 pt/ft at 17 x 11 in | Dimensioned geometry governs

DESIGN BASIS

20 duplex receptacle locations.

Illustrative allowance: 180 VA/location.

C2: R01-R05; C4: R06-R10.

C6: R11-R15; C7: R16-R20.

Device tags show ID / circuit number.

Dedicated equipment is located conceptually.

EQ01 refrigerator: 600 VA.

EQ02 microwave: 1,500 VA.

EQ03 dishwasher: 1,200 VA.

EQ04 HVAC terminal: 2,400 VA, 120 V.

EQ05 exhaust fan: 120 VA.

TO VERIFY

Actual nameplate ratings and supply voltage.

Wet-area protection, mounting, accessibility, receptacle spacing and circuit requirements.

Panel access / working clearances and feeder route require architectural coordination.

CONCEPTUAL DESIGN - NOT FOR CONSTRUCTION

Python-generated portfolio study. Not reviewed in AutoCAD; no code compliance, professional seal, or permit suitability claimed.

E-102

PRELIMINARY PANEL SCHEDULE & LOAD SUMMARY

AMARACHI | COMMERCIAL OFFICE STUDY | 1,200 SQ. FT. | REV 01 | 2026-11-06

LP-1 | Assumed 120/208 V, 3-phase, 4-wire wye | All listed branches are assumed 120 V line-to-neutral.

12 conceptual single-pole circuit positions. No breaker, conductor, bus rating or feeder size selected.

CIRCUIT	PHASE	DESCRIPTION	DEVICE IDS	CONNECTED VA	EST. A @ 120 V
1	A	Lighting: open office	L01-L06	240	2.00
2	A	Open office receptacles	R01-R05	900	7.50
3	B	Lighting: reception + conference	L07-L12	240	2.00
4	B	Open office receptacles	R06-R10	900	7.50
5	C	Lighting: kitchen + washroom	L13-L15	120	1.00
6	C	Reception / conference receptacles	R11-R15	900	7.50
7	A	Kitchen / washroom receptacles	R16-R20	900	7.50
8	A	Refrigerator - assumed	EQ01	600	5.00
9	B	Microwave - assumed	EQ02	1500	12.50
10	B	Dishwasher - assumed	EQ03	1200	10.00
11	C	HVAC terminal - assumed 120 V	EQ04	2400	20.00
12	C	Washroom exhaust - assumed	EQ05	120	1.00

Total connected load: 10,020 VA = 10.02 kVA. Illustrative demand factor: 100%; demand: 10,020 VA.

Phase A: 2,640 VA / 22.0 A | B: 3,840 VA / 32.0 A | C: 3,540 VA / 29.5 A

Phase allocation is preliminary and unbalanced. Maximum phase current = 32.0 A; rebalancing remains open.

Do not use S / (sqrt(3) x 208 V) as the design current for this unbalanced schedule.

Apparent power (VA) is used directly. Real power (kW) cannot be inferred without equipment power factors.

Excluded / unconfirmed: central HVAC, water heating, emergency systems, communications and future loads.

This is a partial tenant-load study, not a completed service calculation. No spare or growth allowance included.

CONCEPTUAL DESIGN - NOT FOR CONSTRUCTION

Python-generated portfolio study. Not reviewed in AutoCAD; no code compliance, professional seal, or permit suitability claimed.

ELECTRICAL LEGEND & DEVICE SCHEDULE

AMARACHI | COMMERCIAL OFFICE STUDY | 1,200 SQ. FT. | REV 01 | 2026-11-06

Lxx: illustrative LED luminaire, 40 VA each

Rxx: duplex receptacle, 180 VA assumed per location

Sx: local lighting switch / control point

EQxx: dedicated equipment location (diagrammatic)

LP-1: conceptual panel location

Dashed: switching association; DXF layer E-CIRCUITS

DEVICE INVENTORY

L01-L15: 15 fixtures / 600 VA.

R01-R20: 20 locations / 3,600 VA.

S1-S5: 5 control points / no added load.

EQ01-EQ05: 5 items / 5,820 VA.

LP-1: 1 panel / loads tabulated on E-103.

DXF LAYERS

A-WALLS, A-DOORS

E-LIGHTING, E-POWER

E-SWITCHES, E-CIRCUITS, E-TEXT

UNITS & USE

DXF model-space units: feet (INSUNITS=2).

Open in AutoCAD and verify 40 x 30 ft.

Model-space annotations are provided.

No native DWG, layouts, or plot styles included.

DXF sheets use the same floor coordinates.

CONCEPTUAL DESIGN - NOT FOR CONSTRUCTION

Python-generated portfolio study. Not reviewed in AutoCAD; no code compliance, professional seal, or permit suitability claimed.

DESIGN BASIS, REVIEW ITEMS & METHODOLOGY

AMARACHI | COMMERCIAL OFFICE STUDY | 1,200 SQ. FT. | REV 01 | 2026-11-06

01 / SCENARIO & BOUNDARY

Fictitious 40 x 30 ft commercial office. Architectural geometry is generated for this study, not surveyed.

Reception: 216 sq. ft.; conference: 144; kitchen: 120; washroom: 60; circulation: 120; open office: 540.

A single 120/208 V wye panel is assumed. Utility supply and base-building distribution are unverified.

02 / CALCULATION METHOD

Connected VA = sum of device quantities x assumed VA per device, plus dedicated equipment VA.

Demand VA = connected VA x user-configured factor (1.00 in this study; not a prescribed code factor).

Branch current = branch VA / 120 V. Phase current = sum of line-to-neutral branch VA on that phase / 120 V.

Circuit numbering convention: positions 1/2=A, 3/4=B, 5/6=C, repeating. All loads assumed single pole.

03 / OPEN ENGINEERING REVIEW ITEMS

Verify occupancy and applicable OESC rules, demand factors, continuous loads, loading and circuit limits.

Confirm equipment nameplates, protection, conductor ampacity, derating, voltage drop, bonding and grounding.

Review fault levels, interrupting ratings, coordination, feeder capacity, neutral loading and harmonics.

Coordinate exits, emergency lighting, fire alarm, accessibility, ceiling services and wet-location protection.

Establish measured photometric performance and lighting control / energy requirements.

Complete circuit rebalancing, tenant/base-building scope, spare capacity and architectural coordination.

04 / AUTHORSHIP & VALIDATION

Created with Python, ezdxf and ReportLab using documented sample assumptions. Automated checks verify device counts, unique identifiers, load totals, phase totals, PDF page count and DXF file integrity.

No independent engineering review or AutoCAD application review has been completed.

05 / REFERENCE FOR SUBSEQUENT CODE REVIEW

Electrical Safety Authority - Ontario Electrical Safety Code information: https://esasafe.com/role/oesc/

Ontario regulation: https://www.ontario.ca/laws/regulation/990164

These references identify the review framework; no clauses or code-derived sizing rules are applied here.